David Stewart

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Education

Georgia Institute of Technology - B.S. in Industrial Engineering | GPA: 3.6 Graduation: May 2028

Stamps President's Scholar - Full-ride scholarship awarded to the top 1% of applicants
Concentration: Analytics & Data Science | **Coursework**: Database Systems, Engineering Optimization, Simulation Analysis & Design, Quality Improvement Methods, Regression & Forecasting

Experience

Automation Program Management Intern May 2026 - August 2026
TSMC (Taiwan Semiconductor Manufacturing Co.) | Intelligent Manufacturing Engineer Intern *Phoenix, AZ*

- Pinpointed undetected automation failures and independently proposed, scoped, and led a technical program auditing **130** automated procedures across **2,200+** tools at the $165B Arizona fab, uncovering a **4–6%** silent failure rate.
- Translated inconsistent procedure templates into detection specs with owners across departments and Taiwan-based teams in Mandarin, writing SQL and Python over **1,000,000+** daily logs with on-call alerts for critical misses.
- Quantified ∼**600 hours/month** of manual work and an estimated **six figures** in scrap (wafers ∼$25K each) by prioritizing fixes that protect wafer flow, then built a reusable audit template keyed to candidate type.
- Shipped a platform-wide **time-saved-per-run KPI** measuring each automation's true value and 4 Power BI dashboards, now in **active use**, after **winning approval** by presenting to Fab 21's head of manufacturing.

Product Management Fellow September 2026 - Present
Microsoft Student Research *Remote*

- Researching AI-powered incident management and on-call tooling for a Microsoft product team across competitive teardowns, market sizing, and discovery.
- Tearing down a competitor platform across **6** evaluation tracks, from on-call alerting to AI investigation agents and pricing, separating verified capability from vendor marketing with sourced evidence.
- Classifying competitor features on a Kano model into must-haves, differentiators, and sacrificial bets, feeding a unified market matrix, go-to-market recommendation, and prototype for the December readout.

Director of Events August 2026 - Present
Product @ GT

Finance Director, Product Consultant September 2024 - Present
180 Degrees Consulting (Client: Pro Bono Partnership of Atlanta)

- Drove a **30%** increase in monthly active users and **20%** in client registration year over year by auditing engagement data and directing a content management system rebuild.

Strategy and Data Analyst September 2024 - Present
Consult Your Community

- Identified **$1M** in idle inventory for an educational distributor by digitizing legacy paper records into a unified inventory database, then delivered a B2B redirection plan.

Projects

OpsAudit | *Python (scikit-learn), SQL, JavaScript, Anthropic API* | dstew2816.github.io/ops-audit Aug 2026 - Present

- Built an operations audit tool on synthetic factory data that rebuilds what each automated procedure should have caught, surfacing a **26.1%** miss rate invisible to the source platform and ranking fixes by cost at risk.
- Trained decision-tree and random-forest classifiers to identify which of **33** check types a messy, multilingual procedure definition monitors; the forest reached **89%** accuracy (tree **80%**) vs **66%** for hand-written rules.
- Added statistical process control (control charts, Cpk/Ppk, run rules), stress-tested on real fab data (UCI SECOM, **1,567** runs), and LLM layers that write detection SQL from procedure documents and group misses by cause.

Leadership and Skills

Leadership: VP of Philanthropy, Delta Chi | CSA Finance Board | Germination Project Fellow
Technical: SQL / MySQL, Python, Power BI, Tableau, Excel, Git / GitHub, Jira, Lucidchart, LLM APIs and evaluation
Program: Cross-functional program coordination, stakeholder management, requirements gathering, risk prioritization, root cause analysis, process mapping, process automation, statistical process control, KPI definition
Languages and Additional: Fluent in English and Mandarin Chinese | 6 years of commercial and film acting experience